

SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

CHENNAI – 602105

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – CASE STUDY

Course Code: DSA0502

CO Assessed: CO4

Weightage:

Course Title: Query Processing for Data Science

Assessment: Tool 1– CASE STUDY

Total Marks:

CO4 Statement: Explore datasets using analytical techniques such as grouping, correlation detection, joining datasets, and extracting meaningful insights.

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Section / Batch: 2024

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Project Mode: Individual Submission

Code of Conduct

I Sandra Sudevan(192424422) (Name / Reg No) certify that this submission is my original work and that I have adhered to all guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature: Sandra sudevan

Question 6 – Weather and Climate Analysis

A weather department provides **Temperature, Rainfall, Humidity, Wind Speed, and Location datasets**. Explore and integrate the data to identify seasonal patterns, correlations, extreme weather values, and monthly/yearly trends. Present the findings using charts and suitable geographical visualizations.

1. Data Importing & Dataset Description

Import the **Temperature, Rainfall, Humidity, Wind Speed, and Location** datasets using Python/Pandas.

Dataset	Important Columns
Temperature	Date, Location, Temperature
Rainfall	Date, Location, Rainfall
Humidity	Date, Location, Humidity
Wind Speed	Date, Location, Wind Speed
Location	Location, Latitude, Longitude

Objective: Integrate all datasets based on **Date and Location**.

2. Data Exploration

- Display first 5 and last 5 records.
- Check number of rows and columns.
- Check missing values.
- Check duplicate records.
- Check data types.
- Calculate mean, minimum, maximum and standard deviation.
- Check unusual temperature, rainfall and wind-speed values.

3. Joining Multiple Datasets

Join the datasets using:

Common columns: Date + Location

The final integrated dataset contains:

Date | Location | Temperature | Rainfall | Humidity | Wind Speed | Latitude | Longitude

4. Grouping & Aggregation

Calculate:

- Average temperature by month.
- Total rainfall by month.
- Average humidity by location.
- Average wind speed by location.
- Maximum temperature by year.
- Total rainfall by year.

5. Data Segmentation

Divide weather data into:

Seasons:

- Winter
- Summer
- Monsoon
- Post-Monsoon

Temperature categories:

- Cold
- Moderate
- Hot
- Extreme Hot

Rainfall categories:

- Low
- Moderate
- Heavy
- Extreme

6. Correlation Analysis

Study relationships between:

- Temperature ↔ Humidity
- Temperature ↔ Rainfall
- Temperature ↔ Wind Speed
- Humidity ↔ Rainfall

Use a **correlation matrix/heatmap**.

7. Outlier Detection

Use the **IQR method** or **box plots** to identify:

- Extremely high temperatures.
- Extremely low temperatures.
- Abnormally high rainfall.
- Unusually high wind speeds.

- Abnormal humidity values.

8. Time-Based Analysis

Analyze:

- Daily temperature changes.
- Monthly rainfall patterns.
- Seasonal temperature variations.
- Yearly temperature trends.
- Yearly rainfall trends.
- Monthly humidity patterns.

9. Visualization

Suitable charts:

Analysis	Visualization
Temperature trend	Line chart
Monthly rainfall	Bar chart
Temperature distribution	Histogram
Weather outliers	Box plot
Weather correlations	Heatmap
Seasonal comparison	Bar chart
Location comparison	Bar chart
Weather by geographical location	Map
Yearly trends	Line chart

10. Key Findings, Conclusion & Recommendations

Key Findings:

- Identify the hottest and coldest locations.
- Determine the months with maximum rainfall.
- Identify seasonal weather patterns.
- Determine whether temperature and humidity have a negative/positive correlation.
- Detect extreme rainfall, temperature and wind-speed events.
- Identify yearly changes in weather conditions.

Conclusion:

The integrated weather dataset helps identify seasonal patterns, relationships between different weather variables, extreme weather conditions and long-term climate trends.

Recommendations:

- Improve weather forecasting during extreme conditions.
- Prepare flood-management plans for heavy-rainfall periods.
- Provide heat alerts during extreme-temperature periods.
- Monitor locations experiencing unusual weather patterns.

Figure 1

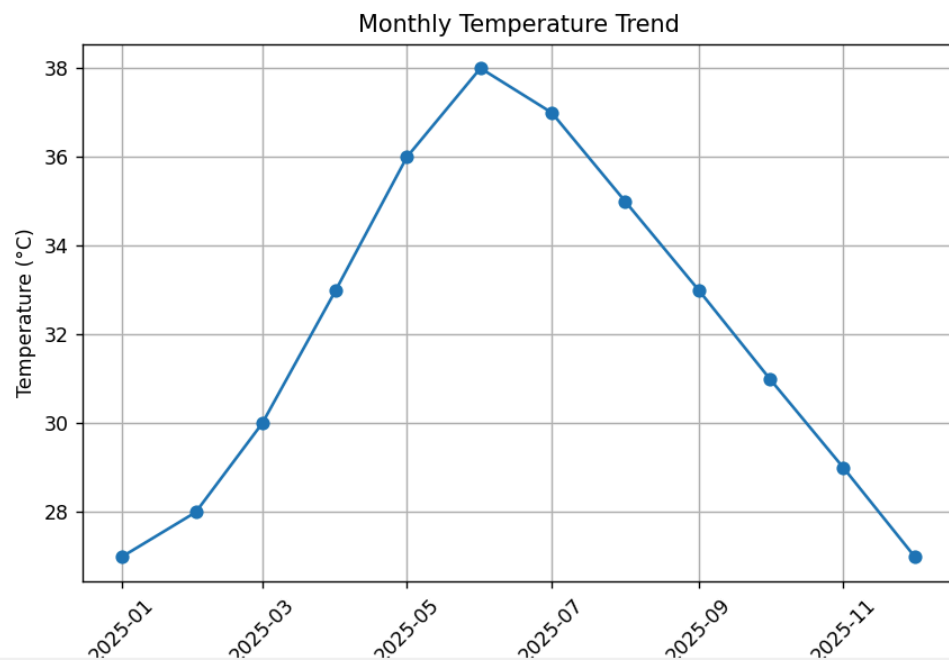


Figure 2

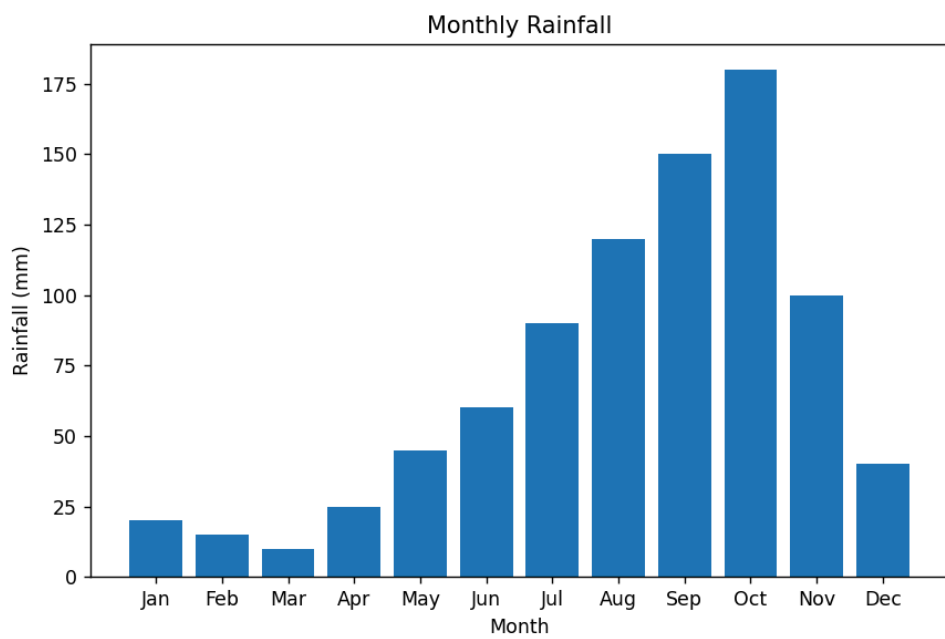


Figure 3

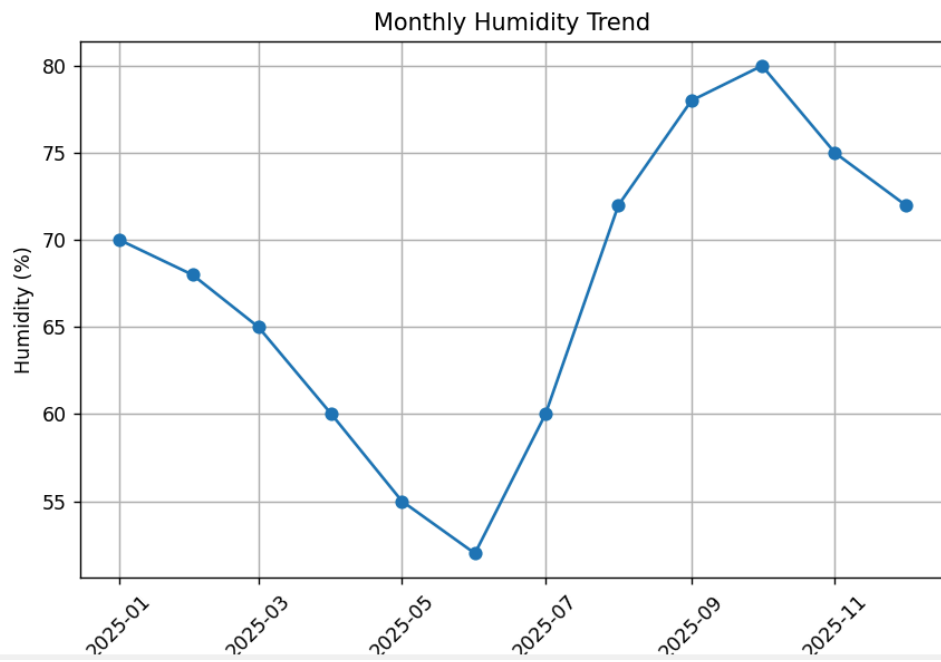


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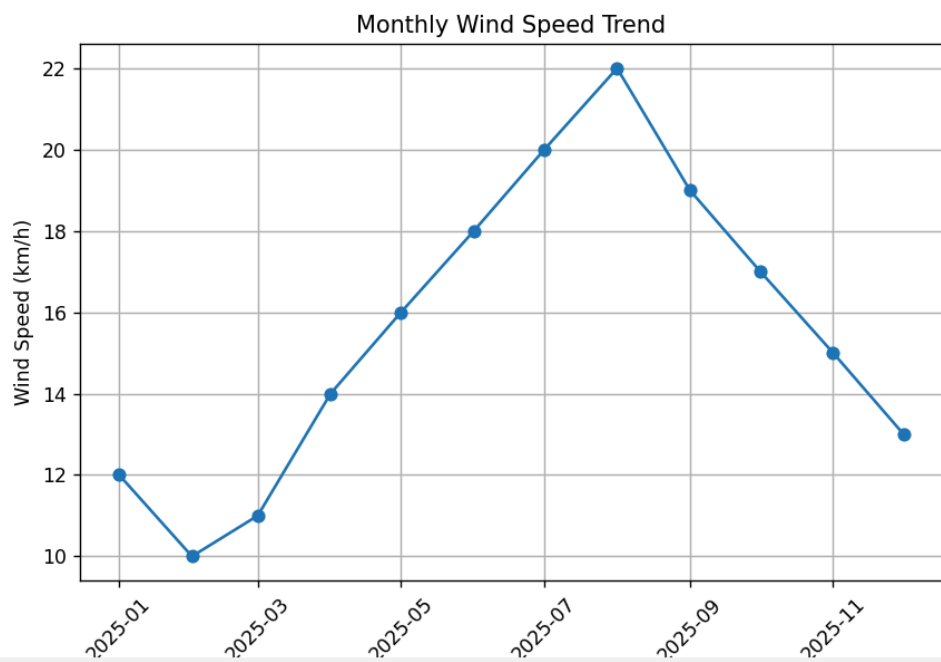


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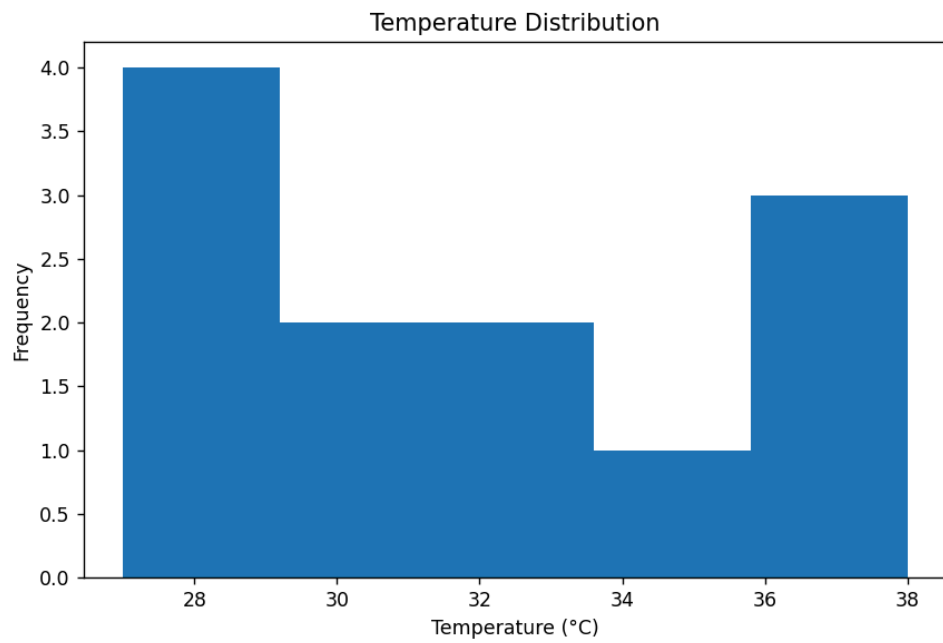
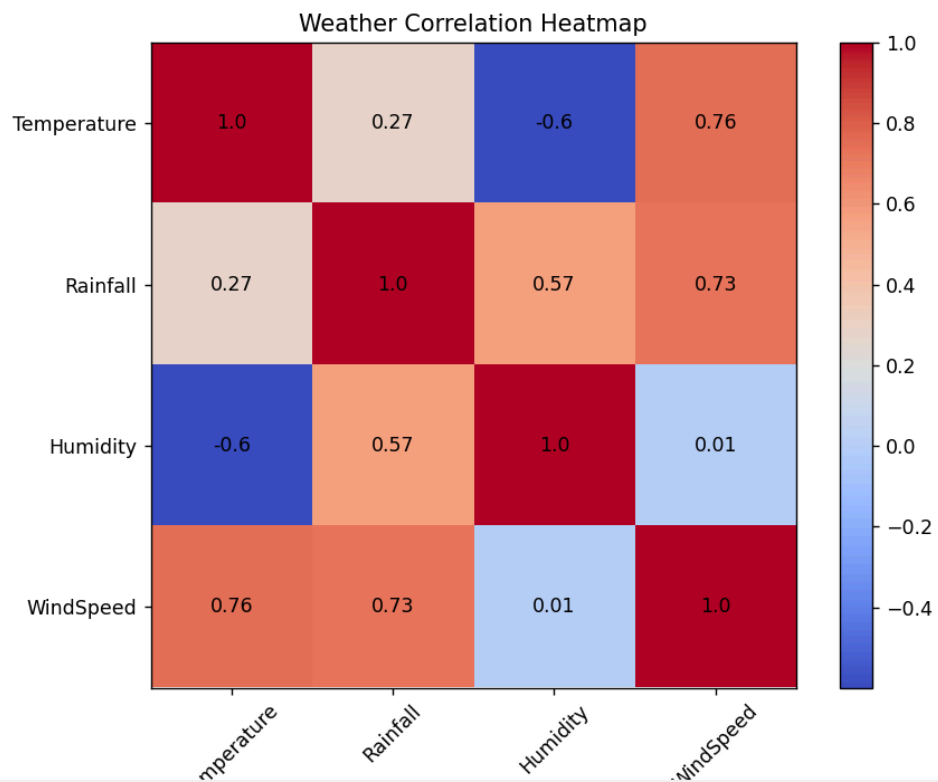
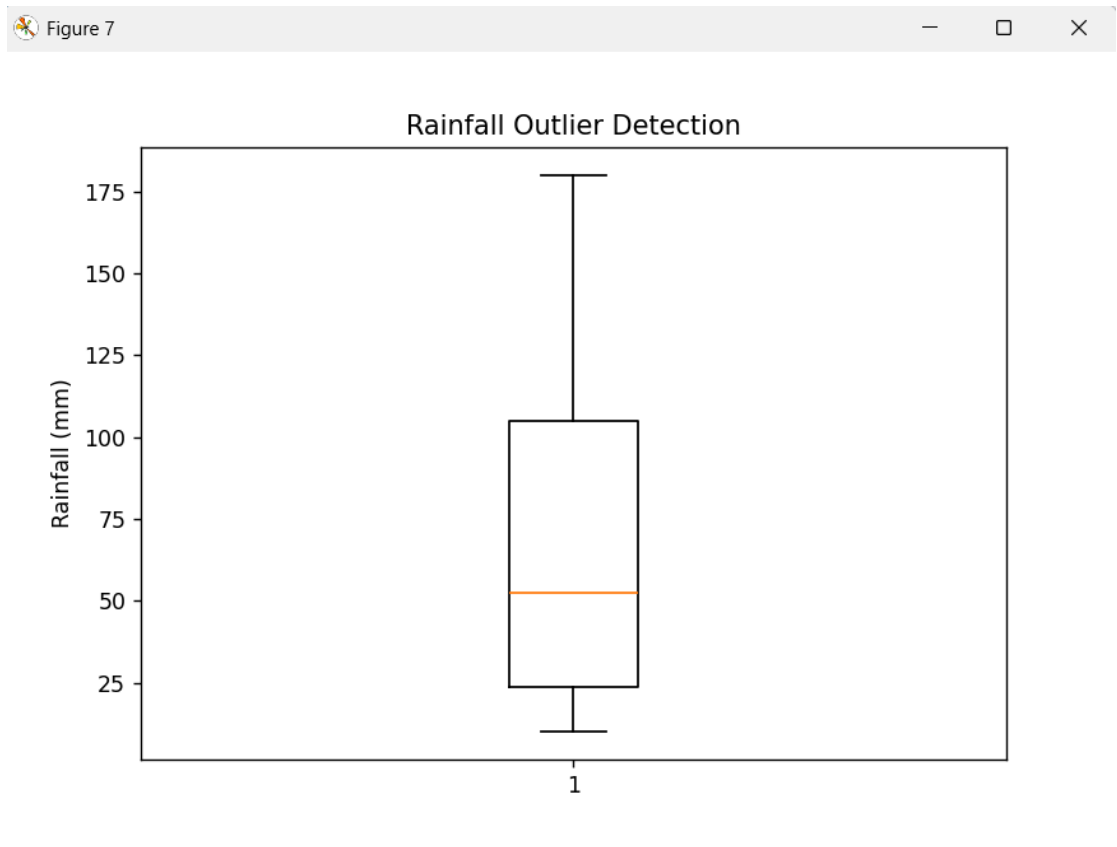


Figure 6





Question 7 – Public Transportation Analysis

A transportation authority provides **Passenger, Route, Schedule, Ticket, and Delay datasets**. Analyze passenger demand, popular routes, peak hours, ticket revenue, and delays. Identify outliers and time-based trends and present the results using charts and maps.

1. Data Importing & Dataset Description

Import the **Passenger, Route, Schedule, Ticket and Delay** datasets.

Dataset	Important Columns
Passenger	Passenger_ID, Date, Route_ID, Passenger_Count
Route	Route_ID, Route_Name, Start, Destination, Distance
Schedule	Schedule_ID, Route_ID, Departure_Time, Arrival_Time

Ticket Ticket_ID, Passenger_ID, Route_ID, Fare

Delay Route_ID, Date, Delay_Minutes

Objective: Integrate these datasets to analyze passenger demand, routes, revenue and delays.

2. Data Exploration

- Display sample records.
- Check dataset dimensions.
- Check missing values.
- Check duplicates.
- Check data types.
- Find minimum and maximum ticket prices.
- Find minimum and maximum delay.
- Examine passenger counts.

3. Joining Multiple Datasets

Join datasets using common identifiers such as:

Passenger_ID, Route_ID, Schedule_ID, and Date.

The final dataset can contain:

Date | Passenger_ID | Route_ID | Route_Name | Departure_Time | Passenger_Count |
Fare | Delay_Minutes

4. Grouping & Aggregation

Calculate:

- Total passengers by route.
- Average passengers per day.
- Total ticket revenue by route.
- Average ticket fare.
- Average delay by route.
- Total delays by month.
- Passenger count by hour.
- Revenue by month/year.

5. Data Segmentation

Segment passengers into:

- Low demand
- Medium demand
- High demand

Segment routes into:

- Low-demand routes
- Medium-demand routes
- High-demand routes

Segment delays into:

- On time: 0 minutes
- Minor delay: 1–10 minutes
- Moderate delay: 11–30 minutes
- Major delay: >30 minutes

6. Correlation Analysis

Analyze relationships between:

- Passenger count ↔ Ticket revenue
- Passenger count ↔ Delay
- Route distance ↔ Fare
- Passenger demand ↔ Delay duration

Use a **correlation matrix and heatmap**.

7. Outlier Detection

Use the **IQR method/box plot** to identify:

- Extremely high passenger counts.
- Unusually high ticket fares.
- Very long delays.
- Routes with abnormal demand.
- Unusual daily revenue.

8. Time-Based Analysis

Analyze:

Peak hours:

- Morning peak
- Afternoon
- Evening peak
- Night

Also analyze:

- Daily passenger demand.
- Weekly passenger demand.
- Monthly passenger demand.
- Monthly ticket revenue.
- Yearly passenger trends.

- Monthly/weekly delays.

9. Visualization

Analysis	Visualization
Passenger demand	Line chart
Popular routes	Bar chart
Peak hours	Bar chart
Ticket revenue	Line/bar chart
Delay distribution	Histogram
Delay outliers	Box plot
Correlation	Heatmap
Route comparison	Bar chart
Route network	Map
Passenger locations/routes	Geographical map

10. Key Findings, Conclusion & Recommendations

Key Findings:

- Identify the most popular routes.
- Identify peak passenger hours.
- Find routes generating the highest revenue.
- Identify routes with frequent delays.
- Determine months with maximum passenger demand.
- Detect unusual passenger counts and delays.
- Examine the relationship between passenger demand and revenue.

Conclusion:

The transportation analysis helps authorities understand passenger demand, route

popularity, peak travel periods, revenue generation and delay patterns.

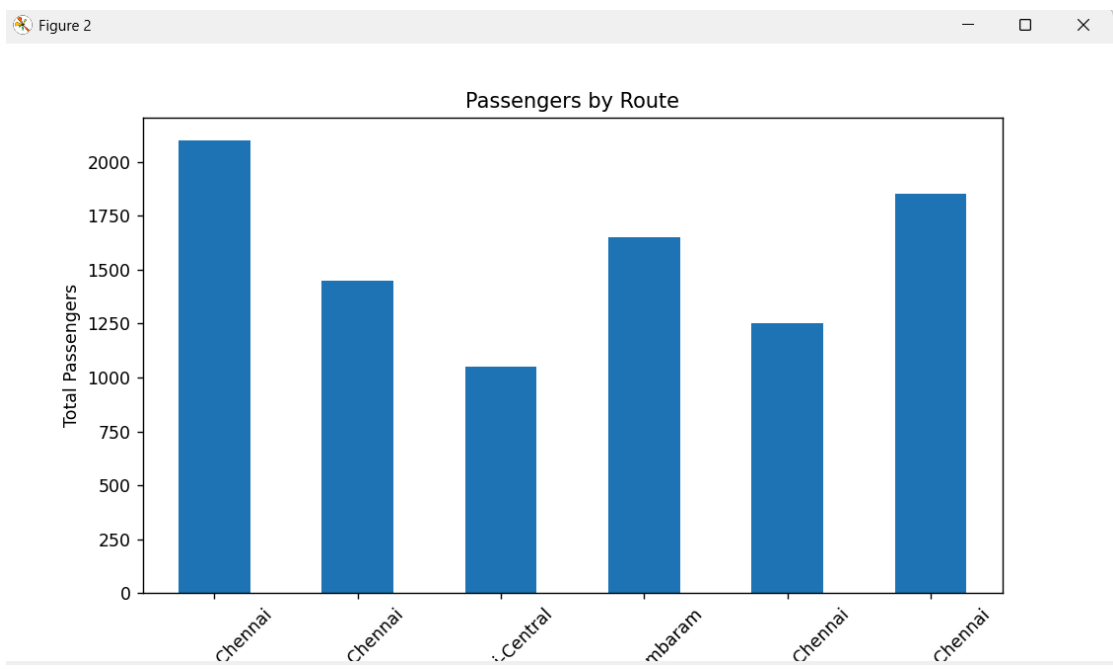
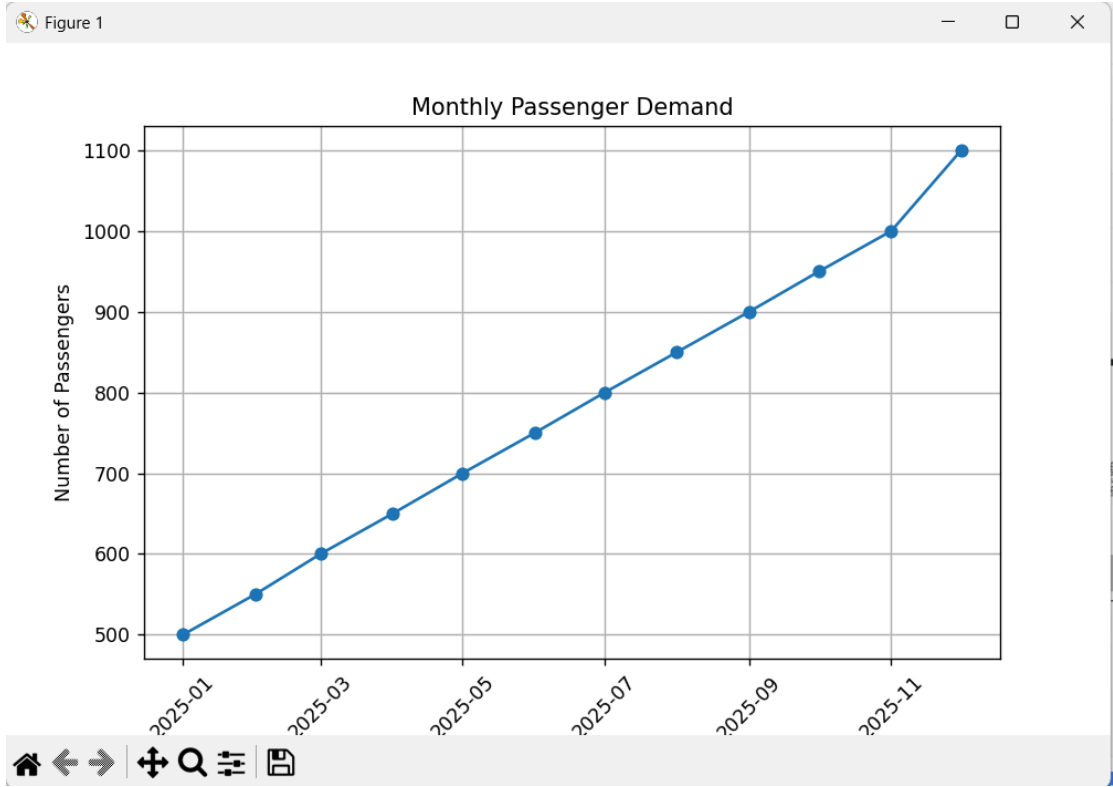


Figure 3

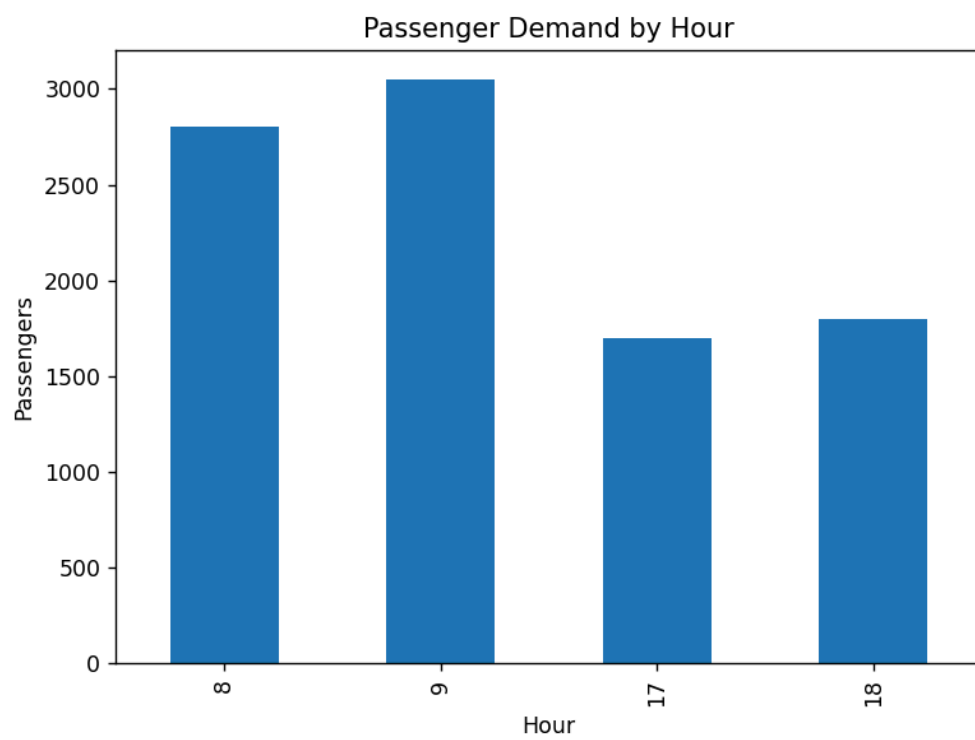


Figure 4

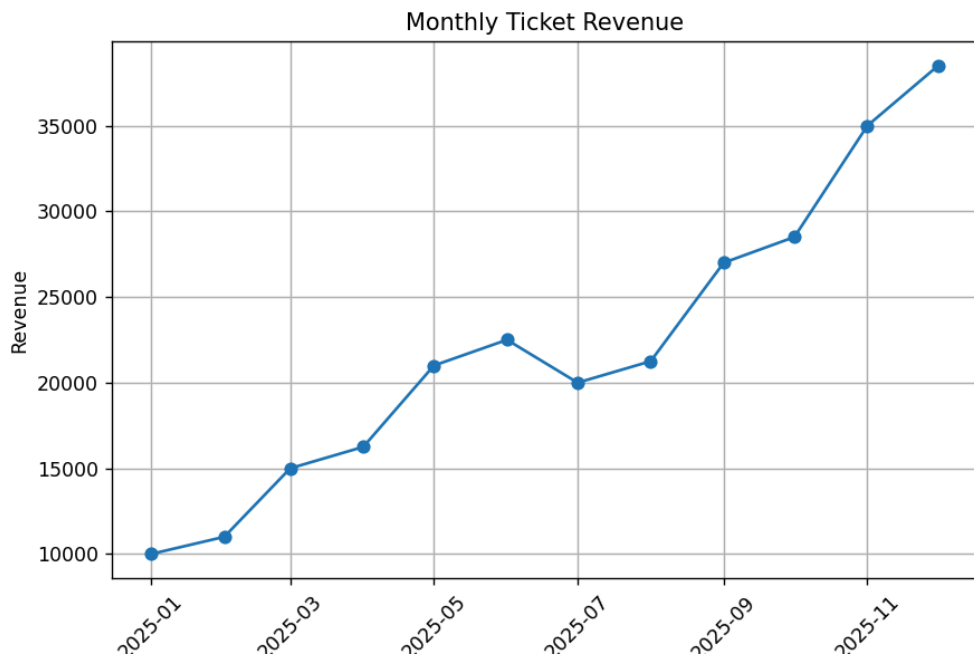


Figure 5

